

Assignment 4 - Final Answers Summary

Exercise 4.1: Global Max/Min

- (i) For $f(x) = \frac{x}{x^2-x+1}$ on $[0, 3]$:
 - Global Maximum: 1, at $x = 1$.
 - Global Minimum: 0, at $x = 0$.
- (ii) For $f(t) = t\sqrt{4-t^2}$ on $[-1, 2]$:
 - Global Maximum: 2, at $t = \sqrt{2}$.
 - Global Minimum: $-\sqrt{3}$, at $t = -1$.
- (iii) For $f(t) = \sqrt[3]{t}(8-t)$ on $[0, 8]$:
 - Global Maximum: $6\sqrt[3]{2}$, at $t = 2$.
 - Global Minimum: 0, at $t = 0$ and $t = 8$.
- (iv) For $f(x) = x - \ln x$ on $[\frac{1}{2}, 2]$:
 - Global Maximum: $2 - \ln 2$, at $x = 2$.
 - Global Minimum: 1, at $x = 1$.

Exercise 4.2: Proof by Differentiation

- (i) $2 \arcsin x = \arccos(1 - 2x^2)$ for $x \geq 0$ is proven.
- (ii) $\arcsin \frac{x-1}{x+1} = 2 \arctan \sqrt{x} - \frac{\pi}{2}$ for $x \geq 0$ is proven.

Exercise 4.3: Inflection Points

The three inflection points of $y = \frac{1+x}{1+x^2}$ are $P_1 = (1, 1)$, $P_2 = (-2 + \sqrt{3}, \frac{1+\sqrt{3}}{4})$, and $P_3 = (-2 - \sqrt{3}, \frac{1-\sqrt{3}}{4})$. It is shown they all lie on the line $y = \frac{1}{4}x + \frac{3}{4}$.

Exercise 4.4: Tangent Curves

It is shown that $y = e^{-x}$ and $y = -e^{-x}$ are tangent to $y = e^{-x} \sin x$ at its inflection points.

Exercise 4.5: Locus of Inflection Points

It is shown that the inflection points (x, y) of $y = x \sin x$ satisfy the equation $y^2(x^2 + 4) = 4x^2$.

Exercise 4.6: L'Hopital's Rule (I)

$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x-h)}{2h} = f'(x)$ is proven.

Exercise 4.7: L'Hopital's Rule (II)

$\lim_{h \rightarrow 0} \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} = f''(x)$ is proven.

Exercise 4.8: Function Derivatives at 0

- (i) $f'(0) = 0$.
- (ii) It is proven by induction that $f^{(k)}(0) = 0$ for all $k \in \mathbb{N}$.

Exercise 4.9: Shortest Distance

Using single-variable calculus optimization, it is shown that the shortest distance from (x_1, y_1) to $Ax + By + C = 0$ is $\frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$.

Exercise 4.10: Limit Evaluation

The value of a that makes $\lim_{x \rightarrow \infty} \left(\frac{x+a}{x-a} \right)^x = e$ true is $a = \frac{1}{2}$.

Exercise 4.11: Ordinary Differential Equations

- (a) For $\ddot{x} + 4\dot{x} + 5x = e^{5t} + e^{-2t} \cos t$:
 - General Solution: $x(t) = e^{-2t}(C_1 \cos t + C_2 \sin t) + \frac{1}{50}e^{5t} + \frac{1}{2}te^{-2t} \sin t$
 - Specific Solution for $x(0) = 1, \dot{x}(0) = 2$:
 $x(t) = e^{-2t} \left(\frac{49}{50} \cos t + \frac{193}{50} \sin t \right) + \frac{1}{50}e^{5t} + \frac{1}{2}te^{-2t} \sin t$

- **(b)** For $\ddot{x} + 4\dot{x} + 4x = e^{-2t}$:
 - General Solution: $x(t) = \left(C_1 + C_2t + \frac{1}{2}t^2\right) e^{-2t}$

Assignment 4 - Detailed Solutions

Exercise 4.1 (40 pts)

To find the global maximum and minimum values of a continuous function f on a closed interval $[a, b]$:

1. Find the critical points of f in the open interval (a, b) .
2. Evaluate f at all critical points and at the endpoints a and b .
3. The largest value is the global maximum, and the smallest value is the global minimum.

(i) (10 pts) $f(x) = \frac{x}{x^2 - x + 1}$; $x \in [0, 3]$

1. Find the derivative: Using the Quotient Rule:

$$f'(x) = \frac{(1)(x^2 - x + 1) - x(2x - 1)}{(x^2 - x + 1)^2}$$

$$f'(x) = \frac{x^2 - x + 1 - 2x^2 + x}{(x^2 - x + 1)^2} = \frac{1 - x^2}{(x^2 - x + 1)^2}$$

2. Find critical points: Set $f'(x) = 0$, which means $1 - x^2 = 0$. This gives $x = 1$ or $x = -1$. The only critical point in the interval $[0, 3]$ is $x = 1$. **3. Compare values:**

- Endpoint $x = 0$: $f(0) = \frac{0}{1} = 0$
- Critical Point $x = 1$: $f(1) = \frac{1}{1-1+1} = 1$
- Endpoint $x = 3$: $f(3) = \frac{3}{9-3+1} = \frac{3}{7}$

Comparing 0, 1, and $\frac{3}{7}$:

- **Global Maximum:** 1 (at $x = 1$)
- **Global Minimum:** 0 (at $x = 0$)

(ii) (10 pts) $f(t) = t\sqrt{4-t^2}$, $t \in [-1, 2]$

1. **Find the derivative:** Using the Product Rule:

$$f'(t) = (1)\sqrt{4-t^2} + t \left(\frac{1}{2\sqrt{4-t^2}} \cdot (-2t) \right)$$

$$f'(t) = \sqrt{4-t^2} - \frac{t^2}{\sqrt{4-t^2}} = \frac{(4-t^2) - t^2}{\sqrt{4-t^2}} = \frac{4-2t^2}{\sqrt{4-t^2}}$$

2. **Find critical points:** Critical points occur when $f'(t) = 0$ or $f'(t)$ is undefined.

- $f'(t) = 0$: $4 - 2t^2 = 0 \implies t^2 = 2 \implies t = \sqrt{2}$ or $t = -\sqrt{2}$.
- $f'(t)$ undefined: $4 - t^2 = 0 \implies t = 2$ or $t = -2$.

The only critical point in the open interval $(-1, 2)$ is $t = \sqrt{2}$. 3. **Compare values:**

- Endpoint $t = -1$: $f(-1) = (-1)\sqrt{4-1} = -\sqrt{3}$
- Critical Point $t = \sqrt{2}$: $f(\sqrt{2}) = \sqrt{2}\sqrt{4-2} = \sqrt{2}\sqrt{2} = 2$
- Endpoint $t = 2$: $f(2) = 2\sqrt{4-4} = 0$

Comparing $-\sqrt{3}$, 2, and 0:

- **Global Maximum:** 2 (at $t = \sqrt{2}$)
- **Global Minimum:** $-\sqrt{3}$ (at $t = -1$)

(iii) (10 pts) $f(t) = \sqrt[3]{t}(8-t) = 8t^{1/3} - t^{4/3}$, $t \in [0, 8]$

1. **Find the derivative:**

$$f'(t) = 8 \cdot \frac{1}{3}t^{-2/3} - \frac{4}{3}t^{1/3} = \frac{8-4t}{3t^{2/3}}$$

2. **Find critical points:**

- $f'(t) = 0$: $8 - 4t = 0 \implies t = 2$.
- $f'(t)$ undefined: $3t^{2/3} = 0 \implies t = 0$.

The critical point in $(0, 8)$ is $t = 2$. The point $t = 0$ is an endpoint. **3. Compare values:**

- Endpoint/Critical Point $t = 0$: $f(0) = 0$
- Critical Point $t = 2$: $f(2) = \sqrt[3]{2}(8 - 2) = 6\sqrt[3]{2}$
- Endpoint $t = 8$: $f(8) = \sqrt[3]{8}(8 - 8) = 0$

- **Global Maximum:** $6\sqrt[3]{2}$ (at $t = 2$)
- **Global Minimum:** 0 (at $t = 0$ and $t = 8$)

(iv) (10 pts) $f(x) = x - \ln x$, $x \in [\frac{1}{2}, 2]$

(Assuming $\log x = \ln x$) **1. Find the derivative:**

$$f'(x) = 1 - \frac{1}{x}$$

2. Find critical points: Set $f'(x) = 0$:

$$1 - \frac{1}{x} = 0 \implies x = 1$$

The critical point $x = 1$ is in the interval $[\frac{1}{2}, 2]$. **3. Compare values:**

- Endpoint $x = 1/2$: $f(1/2) = \frac{1}{2} - \ln(\frac{1}{2}) = \frac{1}{2} + \ln 2 \approx 1.193$
- Critical Point $x = 1$: $f(1) = 1 - \ln(1) = 1$
- Endpoint $x = 2$: $f(2) = 2 - \ln 2 \approx 1.307$

Comparing the values:

- **Global Maximum:** $2 - \ln 2$ (at $x = 2$)
 - **Global Minimum:** 1 (at $x = 1$)
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Exercise 4.2 (20 pts)

(i) (10 pts) $2 \arcsin x = \arccos(1 - 2x^2)$, $x \geq 0$

1. Differentiate both sides: Let $f(x) = 2 \arcsin x$.

$$f'(x) = 2 \cdot \frac{1}{\sqrt{1-x^2}} = \frac{2}{\sqrt{1-x^2}}$$

Let $g(x) = \arccos(1 - 2x^2)$. Using the Chain Rule:

$$g'(x) = \frac{-1}{\sqrt{1-(1-2x^2)^2}} \cdot (-4x)$$

$$g'(x) = \frac{4x}{\sqrt{1-(1-4x^2+4x^4)}} = \frac{4x}{\sqrt{4x^2-4x^4}}$$

$$g'(x) = \frac{4x}{\sqrt{4x^2(1-x^2)}} = \frac{4x}{2\sqrt{x^2}\sqrt{1-x^2}}$$

Since $x \geq 0$, we have $\sqrt{x^2} = x$.

$$g'(x) = \frac{4x}{2x\sqrt{1-x^2}} = \frac{2}{\sqrt{1-x^2}}$$

2. Compare derivatives and check a point: Since $f'(x) = g'(x)$, we know $f(x) = g(x) + C$. We check a point in the domain, $x = 0$:

- $f(0) = 2 \arcsin(0) = 0$
- $g(0) = \arccos(1 - 0) = \arccos(1) = 0$

Since $f(0) = g(0)$, the constant $C = 0$. Thus, $2 \arcsin x = \arccos(1 - 2x^2)$ for $x \geq 0$.

(ii) (10 pts) $\arcsin \frac{x-1}{x+1} = 2 \arctan \sqrt{x} - \frac{\pi}{2}$

This identity is for $x \geq 0$. **1. Differentiate both sides:** Let $f(x) = \arcsin \frac{x-1}{x+1}$.

$$f'(x) = \frac{1}{\sqrt{1-\left(\frac{x-1}{x+1}\right)^2}} \cdot \left(\frac{(1)(x+1) - (x-1)(1)}{(x+1)^2} \right)$$

$$f'(x) = \frac{1}{\sqrt{\frac{(x+1)^2-(x-1)^2}{(x+1)^2}}} \cdot \left(\frac{2}{(x+1)^2} \right)$$

$$f'(x) = \frac{1}{\sqrt{\frac{4x}{(x+1)^2}}} \cdot \frac{2}{(x+1)^2} = \frac{|x+1|}{2\sqrt{x}} \cdot \frac{2}{(x+1)^2}$$

Since $x \geq 0$, $x+1 > 0$, so $|x+1| = x+1$.

$$f'(x) = \frac{x+1}{2\sqrt{x}} \cdot \frac{2}{(x+1)^2} = \frac{1}{\sqrt{x}(x+1)}$$

Now, let $g(x) = 2 \arctan \sqrt{x} - \frac{\pi}{2}$.

$$g'(x) = 2 \cdot \frac{1}{1 + (\sqrt{x})^2} \cdot \left(\frac{1}{2\sqrt{x}} \right) - 0$$

$$g'(x) = \frac{2}{1+x} \cdot \left(\frac{1}{2\sqrt{x}} \right) = \frac{1}{(1+x)\sqrt{x}}$$

2. Compare derivatives and check a point: Since $f'(x) = g'(x)$, we know $f(x) = g(x) + C$. We check a convenient point, $x = 1$:

- $f(1) = \arcsin\left(\frac{1-1}{1+1}\right) = \arcsin(0) = 0$
- $g(1) = 2 \arctan(\sqrt{1}) - \frac{\pi}{2} = 2 \arctan(1) - \frac{\pi}{2} = 2\left(\frac{\pi}{4}\right) - \frac{\pi}{2} = 0$

Since $f(1) = g(1)$, we have $C = 0$. The identity is proven.

Exercise 4.3 (10 pts)

Show that $y = \frac{1+x}{1+x^2}$ has three inflection points, and they lie on one line. **1. Find y' and y'' :**

$$y' = \frac{(1)(1+x^2) - (1+x)(2x)}{(1+x^2)^2} = \frac{1-2x-x^2}{(1+x^2)^2}$$

$$y'' = \frac{(-2-2x)(1+x^2)^2 - (1-2x-x^2) \cdot 2(1+x^2)(2x)}{(1+x^2)^4}$$

Divide by $(1+x^2)$:

$$y'' = \frac{(-2-2x)(1+x^2) - 4x(1-2x-x^2)}{(1+x^2)^3}$$

$$y'' = \frac{(-2-2x^2-2x-2x^3) - (4x-8x^2-4x^3)}{(1+x^2)^3}$$

$$y'' = \frac{2x^3+6x^2-6x-2}{(1+x^2)^3} = \frac{2(x^3+3x^2-3x-1)}{(1+x^2)^3}$$

2. Find inflection points ($y'' = 0$): We must solve $x^3 + 3x^2 - 3x - 1 = 0$. By inspection, $x = 1$ is a root. Factoring out $(x - 1)$:

$$(x - 1)(x^2 + 4x + 1) = 0$$

One root is $x_1 = 1$. The other roots are from $x^2 + 4x + 1 = 0$, using the quadratic formula:

$$x = \frac{-4 \pm \sqrt{16 - 4}}{2} = \frac{-4 \pm \sqrt{12}}{2} = -2 \pm \sqrt{3}$$

So, $x_2 = -2 + \sqrt{3}$ and $x_3 = -2 - \sqrt{3}$. There are three inflection points. **3. Find the coordinates:**

- P_1 at $x_1 = 1$: $y_1 = \frac{1+1}{1+1} = 1$. So $P_1 = (1, 1)$.
- For x_2 and x_3 , we know $x^2 + 4x + 1 = 0 \implies 1 + x^2 = -4x$. P_2 at $x_2 = -2 + \sqrt{3}$:

$$y_2 = \frac{1 + x_2}{1 + x_2^2} = \frac{1 + (-2 + \sqrt{3})}{-4(-2 + \sqrt{3})} = \frac{-1 + \sqrt{3}}{8 - 4\sqrt{3}}$$

$$y_2 = \frac{-1 + \sqrt{3}}{4(2 - \sqrt{3})} \cdot \frac{2 + \sqrt{3}}{2 + \sqrt{3}} = \frac{-2 - \sqrt{3} + 2\sqrt{3} + 3}{4(4 - 3)} = \frac{1 + \sqrt{3}}{4}$$

$$\text{So } P_2 = \left(-2 + \sqrt{3}, \frac{1 + \sqrt{3}}{4}\right).$$

- P_3 at $x_3 = -2 - \sqrt{3}$:

$$y_3 = \frac{1 + x_3}{1 + x_3^2} = \frac{1 + (-2 - \sqrt{3})}{-4(-2 - \sqrt{3})} = \frac{-1 - \sqrt{3}}{8 + 4\sqrt{3}}$$

$$y_3 = \frac{-(1 + \sqrt{3})}{4(2 + \sqrt{3})} \cdot \frac{2 - \sqrt{3}}{2 - \sqrt{3}} = \frac{-(2 - \sqrt{3} + 2\sqrt{3} - 3)}{4(4 - 3)} = \frac{-(-1 + \sqrt{3})}{4} = \frac{1 - \sqrt{3}}{4}$$

$$\text{So } P_3 = \left(-2 - \sqrt{3}, \frac{1 - \sqrt{3}}{4}\right).$$

4. Check for collinearity: The slope between P_1 and P_2 :

$$m_{12} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\frac{1 + \sqrt{3}}{4} - 1}{-2 + \sqrt{3} - 1} = \frac{\frac{-3 + \sqrt{3}}{4}}{-3 + \sqrt{3}} = \frac{1}{4}$$

The slope between P_2 and P_3 :

$$m_{23} = \frac{y_3 - y_2}{x_3 - x_2} = \frac{\frac{1 - \sqrt{3}}{4} - \frac{1 + \sqrt{3}}{4}}{(-2 - \sqrt{3}) - (-2 + \sqrt{3})} = \frac{\frac{-2\sqrt{3}}{4}}{-2\sqrt{3}} = \frac{1}{4}$$

Since the slopes are equal, $m_{12} = m_{23}$, the three points lie on the same line.

Exercise 4.4 (10 pts)

Show that $y = e^{-x}$ and $y = -e^{-x}$ touch $y = e^{-x} \sin x$ at its inflection points. Let $f(x) = e^{-x} \sin x$, $g(x) = e^{-x}$, and $h(x) = -e^{-x}$. **1. Find inflection points of $f(x)$:**

$$f'(x) = -e^{-x} \sin x + e^{-x} \cos x = e^{-x}(\cos x - \sin x)$$

$$f''(x) = -e^{-x}(\cos x - \sin x) + e^{-x}(-\sin x - \cos x) = -2e^{-x} \cos x$$

Inflection points occur when $f''(x) = 0$, which means $\cos x = 0$. This occurs when $x = \frac{\pi}{2} + n\pi$ for any integer n . **2. Check tangency at these points: Case 1: n is even ($n = 2k$)**
 $x = \frac{\pi}{2} + 2k\pi$. At these points, $\sin x = 1$ and $\cos x = 0$.

- **Compare y-values:** $f(x) = e^{-x} \sin x = e^{-x}(1) = e^{-x}$ $g(x) = e^{-x}$ The values match: $f(x) = g(x)$.
- **Compare derivatives:** $f'(x) = e^{-x}(\cos x - \sin x) = e^{-x}(0 - 1) = -e^{-x}$ $g'(x) = \frac{d}{dx}(e^{-x}) = -e^{-x}$ The derivatives match: $f'(x) = g'(x)$.

Thus, $f(x)$ is tangent to $g(x) = e^{-x}$ at these points. **Case 2: n is odd ($n = 2k + 1$)**
 $x = \frac{\pi}{2} + (2k + 1)\pi = \frac{3\pi}{2} + 2k\pi$. At these points, $\sin x = -1$ and $\cos x = 0$.

- **Compare y-values:** $f(x) = e^{-x} \sin x = e^{-x}(-1) = -e^{-x}$ $h(x) = -e^{-x}$ The values match: $f(x) = h(x)$.
- **Compare derivatives:** $f'(x) = e^{-x}(\cos x - \sin x) = e^{-x}(0 - (-1)) = e^{-x}$ $h'(x) = \frac{d}{dx}(-e^{-x}) = e^{-x}$ The derivatives match: $f'(x) = h'(x)$.

Thus, $f(x)$ is tangent to $h(x) = -e^{-x}$ at these points.

Exercise 4.5 (10 pts)

Show that the inflection points of $y = x \sin x$ lie on $y^2(x^2 + 4) = 4x^2$. **1. Find y'' :**

$$y = x \sin x$$

$$y' = \sin x + x \cos x$$

$$y'' = \cos x + (\cos x - x \sin x) = 2 \cos x - x \sin x$$

2. Set up equations for inflection points: An inflection point (x, y) must satisfy $y = x \sin x$ and $y'' = 0$.

1. $y = x \sin x$

$$2. \quad 2 \cos x - x \sin x = 0$$

3. Eliminate trigonometric functions: From (1), we solve for $\sin x$ (assuming $x \neq 0$, as $x = 0$ is not an inflection point):

$$\sin x = \frac{y}{x}$$

Substitute this into (2):

$$2 \cos x - x \left(\frac{y}{x} \right) = 0 \implies 2 \cos x = y \implies \cos x = \frac{y}{2}$$

Use the identity $\sin^2 x + \cos^2 x = 1$:

$$\begin{aligned} \left(\frac{y}{x} \right)^2 + \left(\frac{y}{2} \right)^2 &= 1 \\ \frac{y^2}{x^2} + \frac{y^2}{4} &= 1 \end{aligned}$$

Multiply by $4x^2$ to clear denominators:

$$4y^2 + x^2y^2 = 4x^2$$

Factor out y^2 :

$$y^2(4 + x^2) = 4x^2$$

This is $y^2(x^2 + 4) = 4x^2$, as required.

Exercise 4.6 (10 pts)

If f' is continuous, use L'Hopital's Rule to show that

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x-h)}{2h} = f'(x)$$

This is a limit as $h \rightarrow 0$. **1. Check limit form:** As $h \rightarrow 0$, the numerator $f(x+h) - f(x-h) \rightarrow f(x) - f(x) = 0$. The denominator $2h \rightarrow 0$. This is a $\frac{0}{0}$ indeterminate form. **2. Apply L'Hopital's Rule:** We differentiate the numerator and denominator with respect to h .

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(x+h) - f(x-h)}{2h} &\stackrel{L'H}{=} \lim_{h \rightarrow 0} \frac{\frac{d}{dh}(f(x+h) - f(x-h))}{\frac{d}{dh}(2h)} \\ &= \lim_{h \rightarrow 0} \frac{f'(x+h) \cdot (1) - f'(x-h) \cdot (-1)}{2} \\ &= \lim_{h \rightarrow 0} \frac{f'(x+h) + f'(x-h)}{2} \end{aligned}$$

3. Evaluate the new limit: Since f' is continuous, we can evaluate the limit by direct substitution of $h = 0$:

$$\frac{f'(x+0) + f'(x-0)}{2} = \frac{f'(x) + f'(x)}{2} = \frac{2f'(x)}{2} = f'(x)$$

The statement is proven.

Exercise 4.7 (10 pts)

If f'' is continuous, show that

$$\lim_{h \rightarrow 0} \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} = f''(x)$$

1. Check limit form: As $h \rightarrow 0$, the numerator $f(x+h) - 2f(x) + f(x-h) \rightarrow f(x) - 2f(x) + f(x) = 0$. The denominator $h^2 \rightarrow 0$. This is a $\frac{0}{0}$ indeterminate form. **2. Apply L'Hopital's Rule (First time):** We differentiate w.r.t. h .

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} &\stackrel{L'H}{=} \lim_{h \rightarrow 0} \frac{f'(x+h) \cdot (1) - 0 + f'(x-h) \cdot (-1)}{2h} \\ &= \lim_{h \rightarrow 0} \frac{f'(x+h) - f'(x-h)}{2h} \end{aligned}$$

3. Check new limit form: As $h \rightarrow 0$, the numerator $f'(x+h) - f'(x-h) \rightarrow f'(x) - f'(x) = 0$. The denominator $2h \rightarrow 0$. This is still a $\frac{0}{0}$ form. **4. Apply L'Hopital's Rule (Second time):**

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f'(x+h) - f'(x-h)}{2h} &\stackrel{L'H}{=} \lim_{h \rightarrow 0} \frac{f''(x+h) \cdot (1) - f''(x-h) \cdot (-1)}{2} \\ &= \lim_{h \rightarrow 0} \frac{f''(x+h) + f''(x-h)}{2} \end{aligned}$$

5. Evaluate the final limit: Since f'' is continuous, we can substitute $h = 0$:

$$\frac{f''(x+0) + f''(x-0)}{2} = \frac{f''(x) + f''(x)}{2} = f''(x)$$

The statement is proven.

Exercise 4.8 (20 pts)

$$\text{Let } f(x) = \begin{cases} e^{-1/x}, & \text{if } x > 0 \\ 0, & \text{if } x \leq 0 \end{cases}$$

(i) (10 pts) Compute $f'(0)$.

We must use the limit definition of the derivative:

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{f(h)}{h}$$

We must check the left-hand and right-hand limits. **Left-hand limit** ($h \rightarrow 0^-$): For $h < 0$, $f(h) = 0$.

$$\lim_{h \rightarrow 0^-} \frac{f(h)}{h} = \lim_{h \rightarrow 0^-} \frac{0}{h} = 0$$

Right-hand limit ($h \rightarrow 0^+$): For $h > 0$, $f(h) = e^{-1/h}$.

$$\lim_{h \rightarrow 0^+} \frac{e^{-1/h}}{h}$$

Let $t = 1/h$. As $h \rightarrow 0^+$, $t \rightarrow +\infty$.

$$\lim_{h \rightarrow 0^+} \frac{e^{-1/h}}{h} = \lim_{t \rightarrow \infty} \frac{e^{-t}}{1/t} = \lim_{t \rightarrow \infty} \frac{t}{e^t}$$

This is an $\frac{\infty}{\infty}$ form. We apply L'Hopital's Rule:

$$\lim_{t \rightarrow \infty} \frac{t}{e^t} \stackrel{L'H}{=} \lim_{t \rightarrow \infty} \frac{1}{e^t} = 0$$

Since the left-hand limit (0) and the right-hand limit (0) are equal, the limit exists.

$$f'(0) = 0$$

(ii) (10 pts) Use induction to show $f^{(k)}(0) = 0$ for all $k \in \mathbb{N}$.

1. (Lemma) First, we show that for $x > 0$, $f^{(k)}(x) = P_k(1/x)e^{-1/x}$, where P_k is a polynomial.

- **Base Case (k=1):** $f'(x) = e^{-1/x} \cdot (1/x^2)$. Let $t = 1/x$. Then $f'(x) = t^2 e^{-1/x}$. $P_1(t) = t^2$ is a polynomial.
- **Inductive Step:** Assume $f^{(k)}(x) = P_k(1/x)e^{-1/x}$.

$$\begin{aligned} f^{(k+1)}(x) &= \frac{d}{dx} (P_k(1/x)e^{-1/x}) \\ &= \left(P'_k(1/x) \cdot \frac{-1}{x^2} \right) e^{-1/x} + P_k(1/x) \left(e^{-1/x} \cdot \frac{1}{x^2} \right) \\ &= \frac{1}{x^2} (-P'_k(1/x) + P_k(1/x)) e^{-1/x} \end{aligned}$$

Let $t = 1/x$. The expression $P_{k+1}(t) = t^2(-P'_k(t) + P_k(t))$ is a new polynomial in t .

2. (Main Proof) We prove $f^{(k)}(0) = 0$ by induction.

- **Base Case (k=1):** From part (i), $f'(0) = 0$.
- **Inductive Step:** Assume $f^{(k)}(0) = 0$ for some $k \geq 1$. We must show $f^{(k+1)}(0) = 0$.

$$f^{(k+1)}(0) = \lim_{h \rightarrow 0} \frac{f^{(k)}(h) - f^{(k)}(0)}{h} = \lim_{h \rightarrow 0} \frac{f^{(k)}(h)}{h}$$

Left-limit ($h \rightarrow 0^-$): For $h < 0$, $f(h) = 0$, so $f^{(k)}(h) = 0$.

$$\lim_{h \rightarrow 0^-} \frac{0}{h} = 0$$

Right-limit ($h \rightarrow 0^+$): We use our lemma:

$$\lim_{h \rightarrow 0^+} \frac{P_k(1/h)e^{-1/h}}{h}$$

Let $t = 1/h$. As $h \rightarrow 0^+$, $t \rightarrow \infty$.

$$\lim_{t \rightarrow \infty} \frac{P_k(t)e^{-t}}{1/t} = \lim_{t \rightarrow \infty} \frac{tP_k(t)}{e^t}$$

Let $Q(t) = tP_k(t)$, which is another polynomial. We evaluate $\lim_{t \rightarrow \infty} \frac{Q(t)}{e^t}$. By repeated application of L'Hopital's Rule, this limit is 0.

Since the left and right limits both equal 0, $f^{(k+1)}(0) = 0$. By induction, $f^{(k)}(0) = 0$ for all $k \in \mathbb{N}$.

Exercise 4.9 (10 pts)

Show that the shortest distance from the point (x_1, y_1) to the straight line $Ax + By + C = 0$ is

$$\frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

Justification of Method: We will find the point (x, y) on the line that minimizes its distance to (x_1, y_1) . We can minimize the *squared distance*, S :

$$S(x, y) = (x - x_1)^2 + (y - y_1)^2$$

We use the constraint $Ax + By + C = 0$ to express one variable in terms of the other. Assume $B \neq 0$.

$$y = -\frac{A}{B}x - \frac{C}{B}$$

Substitute this into S to make it a function of a single variable, x :

$$S(x) = (x - x_1)^2 + \left(\left(-\frac{A}{B}x - \frac{C}{B} \right) - y_1 \right)^2$$

To find the minimum, we find the derivative $S'(x)$ and set it to 0.

$$S'(x) = 2(x - x_1) + 2 \left(-\frac{A}{B}x - \frac{C}{B} - y_1 \right) \cdot \left(-\frac{A}{B} \right) = 0$$

Divide by 2 and substitute back $y = -\frac{A}{B}x - \frac{C}{B}$:

$$(x - x_1) - \frac{A}{B}(y - y_1) = 0 \implies B(x - x_1) = A(y - y_1)$$

This gives us the components of the distance vector. Let $K = Ax_1 + By_1 + C$. We can write the line equation as:

$$A(x - x_1 + x_1) + B(y - y_1 + y_1) + C = 0$$

$$A(x - x_1) + B(y - y_1) + (Ax_1 + By_1 + C) = 0 \implies A(x - x_1) + B(y - y_1) = -K$$

We now have a system of two equations:

1. $B(x - x_1) - A(y - y_1) = 0$
2. $A(x - x_1) + B(y - y_1) = -K$

From (1), $y - y_1 = \frac{B}{A}(x - x_1)$. Substitute this into (2):

$$\begin{aligned} A(x - x_1) + B \left(\frac{B}{A}(x - x_1) \right) &= -K \\ (x - x_1) \left(A + \frac{B^2}{A} \right) &= -K \implies (x - x_1) \left(\frac{A^2 + B^2}{A} \right) = -K \\ x - x_1 &= \frac{-AK}{A^2 + B^2} \end{aligned}$$

Similarly, from (1), $x - x_1 = \frac{A}{B}(y - y_1)$. Substitute this into (2):

$$\begin{aligned} A \left(\frac{A}{B}(y - y_1) \right) + B(y - y_1) &= -K \\ (y - y_1) \left(\frac{A^2}{B} + B \right) &= -K \implies (y - y_1) \left(\frac{A^2 + B^2}{B} \right) = -K \\ y - y_1 &= \frac{-BK}{A^2 + B^2} \end{aligned}$$

Now find the squared distance $D^2 = (x - x_1)^2 + (y - y_1)^2$:

$$\begin{aligned} D^2 &= \left(\frac{-AK}{A^2 + B^2} \right)^2 + \left(\frac{-BK}{A^2 + B^2} \right)^2 \\ D^2 &= \frac{A^2 K^2}{(A^2 + B^2)^2} + \frac{B^2 K^2}{(A^2 + B^2)^2} \end{aligned}$$

$$D^2 = \frac{(A^2 + B^2)K^2}{(A^2 + B^2)^2} = \frac{K^2}{A^2 + B^2}$$

Substitute $K = Ax_1 + By_1 + C$ back in:

$$D^2 = \frac{(Ax_1 + By_1 + C)^2}{A^2 + B^2}$$

Taking the square root gives the distance D :

$$D = \sqrt{\frac{(Ax_1 + By_1 + C)^2}{A^2 + B^2}} = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

(This derivation assumes $A \neq 0$ and $B \neq 0$. If $A = 0$ or $B = 0$, the formula still holds).

Exercise 4.10 (10 pts)

For what value of a is the following equation true?

$$\lim_{x \rightarrow \infty} \left(\frac{x+a}{x-a} \right)^x = e$$

Let $L = \lim_{x \rightarrow \infty} \left(\frac{x+a}{x-a} \right)^x$. This is a 1^∞ indeterminate form. We evaluate the limit by first taking the natural logarithm.

$$\ln L = \ln \left[\lim_{x \rightarrow \infty} \left(\frac{x+a}{x-a} \right)^x \right]$$

By continuity of $\ln$:

$$\ln L = \lim_{x \rightarrow \infty} \left[\ln \left(\frac{x+a}{x-a} \right)^x \right] = \lim_{x \rightarrow \infty} x \ln \left(\frac{x+a}{x-a} \right)$$

$$\ln L = \lim_{x \rightarrow \infty} \frac{\ln(x+a) - \ln(x-a)}{1/x}$$

This is a $\frac{0}{0}$ indeterminate form. We apply L'Hopital's Rule:

$$\ln L \stackrel{L'H}{=} \lim_{x \rightarrow \infty} \frac{\frac{1}{x+a} - \frac{1}{x-a}}{-1/x^2}$$

Simplify the numerator:

$$\frac{1}{x+a} - \frac{1}{x-a} = \frac{(x-a) - (x+a)}{(x+a)(x-a)} = \frac{-2a}{x^2 - a^2}$$

Substitute back into the limit:

$$\ln L = \lim_{x \rightarrow \infty} \frac{\frac{-2a}{x^2 - a^2}}{-1/x^2} = \lim_{x \rightarrow \infty} \frac{2ax^2}{x^2 - a^2}$$

Divide numerator and denominator by x^2 :

$$\ln L = \lim_{x \rightarrow \infty} \frac{2a}{1 - a^2/x^2} = \frac{2a}{1 - 0} = 2a$$

We have $\ln L = 2a$. Since $L = e^{\ln L}$, this means $L = e^{2a}$. We are given that we want $L = e = e^1$.

$$e^{2a} = e^1 \implies 2a = 1 \implies a = \frac{1}{2}$$

Exercise 4.11 (20 pts)

(a) $\ddot{x} + 4\dot{x} + 5x = e^{5t} + e^{-2t} \cos t$

1. Find the homogeneous solution x_h : The characteristic equation is $r^2 + 4r + 5 = 0$. Using the quadratic formula:

$$r = \frac{-4 \pm \sqrt{4^2 - 4(1)(5)}}{2} = \frac{-4 \pm \sqrt{16 - 20}}{2} = \frac{-4 \pm \sqrt{-4}}{2} = -2 \pm i$$

The homogeneous solution is $x_h(t) = e^{-2t}(C_1 \cos t + C_2 \sin t)$.

2. Find the particular solution x_p : We use superposition. Guess $x_p = x_{p1} + x_{p2}$.

- For $f_1(t) = e^{5t}$: Guess $x_{p1} = Ae^{5t}$.

$$(25Ae^{5t}) + 4(5Ae^{5t}) + 5(Ae^{5t}) = e^{5t}$$

$$(25A + 20A + 5A)e^{5t} = 50Ae^{5t} = e^{5t} \implies A = \frac{1}{50}$$

So, $x_{p1}(t) = \frac{1}{50}e^{5t}$.

- For $f_2(t) = e^{-2t} \cos t$: This term is part of the homogeneous solution x_h . This is a case of resonance. We must guess $x_{p2} = te^{-2t}(B \cos t + D \sin t)$. Substituting this into the ODE is very long. A known result for $\ddot{x} + ax + bx = e^{\alpha t} \cos(\beta t)$ when $\alpha \pm i\beta$ is a root is that the $t \cdot (\dots)$ terms cancel, leaving the non- t terms. Let's find the derivatives and collect only the $e^{-2t} \cos t$ and $e^{-2t} \sin t$ terms. Let $x_{p2} = te^{-2t}(B \cos t + D \sin t)$. $\dot{x}_{p2} = e^{-2t}(B \cos t + D \sin t) + \dots$ (terms with t) $\ddot{x}_{p2} = -2e^{-2t}(B \cos t + D \sin t) + e^{-2t}(-B \sin t + D \cos t) + \dots$ (terms with t) $\ddot{x}_{p2} = e^{-2t}((D - 2B) \cos t + (-B - 2D) \sin t) + \dots$

Substitute into $\ddot{x} + 4\dot{x} + 5x = e^{-2t} \cos t$:

$$[e^{-2t}((D - 2B) \cos t + \dots)] + 4[e^{-2t}(B \cos t + \dots)] + 5[0 + \dots] = e^{-2t} \cos t$$

The terms with t are guaranteed to sum to zero. We only need to match the coefficients of $e^{-2t} \cos t$ and $e^{-2t} \sin t$.

$$\cos t : (D - 2B) + 4(B) = 1 \implies D + 2B = 1$$

$$\sin t : (-B - 2D) + 4(D) = 0 \implies -B + 2D = 0 \implies B = 2D$$

Substitute $B = 2D$ into the first equation:

$$D + 2(2D) = 1 \implies 5D = 1 \implies D = \frac{1}{5}$$

Then $B = 2D = \frac{2}{5}$.

Wait, my scratchpad calculation gave $A = 0, B = 1/2$. Let me re-check that. $\ddot{x}_{p2} = e^{-2t}[((-4B + 2D) + t(\dots)) \cos t + ((-2B - 4D) + t(\dots)) \sin t]$ (Using B, D) $\dot{x}_{p2} = e^{-2t}[(B + t(\dots)) \cos t + (D + t(\dots)) \sin t]$ $x_{p2} = e^{-2t}[(tB) \cos t + (tD) \sin t]$

Substitute: $\cos t: (-4B + 2D) + 4(B) = 1 \implies 2D = 1 \implies D = 1/2$. $\sin t: (-2B - 4D) + 4(D) = 0 \implies -2B = 0 \implies B = 0$. This matches my previous result ($A = 0, B = 1/2$ where A, B were $\cos, \sin$ coefficients). So, $B = 0$ and $D = 1/2$. $x_{p2}(t) = te^{-2t}(0 \cdot \cos t + \frac{1}{2} \sin t) = \frac{1}{2}te^{-2t} \sin t$.

The full particular solution is $x_p(t) = \frac{1}{50}e^{5t} + \frac{1}{2}te^{-2t} \sin t$.

3. General Solution: $x(t) = x_h(t) + x_p(t)$

$$x(t) = e^{-2t}(C_1 \cos t + C_2 \sin t) + \frac{1}{50}e^{5t} + \frac{1}{2}te^{-2t} \sin t$$

This can be grouped as $x(t) = e^{-2t}(C_1 \cos t + (C_2 + \frac{1}{2}t) \sin t) + \frac{1}{50}e^{5t}$.

4. Find Specific Solution for $x(0) = 1, \dot{x}(0) = 2$: First, $x(0) = 1$:

$$x(0) = e^0(C_1 \cos 0 + (C_2 + 0) \sin 0) + \frac{1}{50}e^0 = 1$$

$$1 \cdot (C_1 \cdot 1 + 0) + \frac{1}{50} = 1 \implies C_1 = 1 - \frac{1}{50} = \frac{49}{50}$$

Now find $\dot{x}(t)$:

$$\dot{x}(t) = -2e^{-2t}(C_1 \cos t + C_2 \sin t + \frac{1}{2}t \sin t) + e^{-2t}(-C_1 \sin t + C_2 \cos t + \frac{1}{2} \sin t + \frac{1}{2}t \cos t) + \frac{5}{50}e^{5t}$$

Now, $\dot{x}(0) = 2$:

$$\dot{x}(0) = -2e^0(C_1 \cdot 1 + 0 + 0) + e^0(-0 + C_2 \cdot 1 + 0 + 0) + \frac{1}{10}e^0 = 2$$

$$-2C_1 + C_2 + \frac{1}{10} = 2$$

Substitute $C_1 = \frac{49}{50}$:

$$\begin{aligned} -2 \left(\frac{49}{50} \right) + C_2 &= 2 - \frac{1}{10} \\ -\frac{49}{25} + C_2 &= \frac{19}{10} \\ C_2 &= \frac{19}{10} + \frac{49}{25} = \frac{19 \cdot 5}{50} + \frac{49 \cdot 2}{50} = \frac{95}{50} + \frac{98}{50} = \frac{193}{50} \end{aligned}$$

The specific solution is:

$$x(t) = e^{-2t} \left(\frac{49}{50} \cos t + \frac{193}{50} \sin t \right) + \frac{1}{50} e^{5t} + \frac{1}{2} t e^{-2t} \sin t$$

or

$$x(t) = e^{-2t} \left(\frac{49}{50} \cos t + \left(\frac{193}{50} + \frac{1}{2} t \right) \sin t \right) + \frac{1}{50} e^{5t}$$

(b) $\ddot{x} + 4\dot{x} + 4x = e^{-2t}$

1. Find the homogeneous solution x_h : The characteristic equation is $r^2 + 4r + 4 = 0$, which is $(r + 2)^2 = 0$. We have a repeated real root $r = -2$. The homogeneous solution is $x_h(t) = C_1 e^{-2t} + C_2 t e^{-2t}$.

2. Find the particular solution x_p : The right-hand side e^{-2t} is part of x_h . $t e^{-2t}$ is also part of x_h . Therefore, we must guess $x_p(t) = A t^2 e^{-2t}$.

$$\begin{aligned} \dot{x}_p &= A(2t e^{-2t} - 2t^2 e^{-2t}) = e^{-2t}(2At - 2At^2) \\ \ddot{x}_p &= -2e^{-2t}(2At - 2At^2) + e^{-2t}(2A - 4At) = e^{-2t}(2A - 8At + 4At^2) \end{aligned}$$

Substitute into the ODE:

$$e^{-2t}(2A - 8At + 4At^2) + 4(e^{-2t}(2At - 2At^2)) + 4(At^2 e^{-2t}) = e^{-2t}$$

Divide by e^{-2t} :

$$(2A - 8At + 4At^2) + (8At - 8At^2) + (4At^2) = 1$$

Group terms by powers of t :

$$t^2(4A - 8A + 4A) + t(-8A + 8A) + (2A) = 1$$

$$0 + 0 + 2A = 1 \implies A = \frac{1}{2}$$

So, $x_p(t) = \frac{1}{2} t^2 e^{-2t}$.

3. General Solution: $x(t) = x_h(t) + x_p(t)$

$$x(t) = C_1 e^{-2t} + C_2 t e^{-2t} + \frac{1}{2} t^2 e^{-2t}$$

$$x(t) = \left(C_1 + C_2 t + \frac{1}{2} t^2 \right) e^{-2t}$$